

Language Technology

Chapter 3: Corpus Processing Tools

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Corpora

A corpus is a collection of texts (written or spoken) or speech recordings. Corpora are typically balanced across different sources: news, novels, etc.

	English	French	German
Most frequent words in a collection of contemporary running texts	<i>the</i>	<i>de</i>	<i>der</i>
	<i>of</i>	<i>le</i> (article)	<i>die</i>
	<i>to</i>	<i>la</i> (article)	<i>und</i>
	<i>in</i>	<i>et</i>	<i>in</i>
	<i>and</i>	<i>les</i>	<i>des</i>
Most frequent words in Genesis	<i>and</i>	<i>et</i>	<i>und</i>
	<i>the</i>	<i>de</i>	<i>die</i>
	<i>of</i>	<i>la</i>	<i>der</i>
	<i>his</i>	<i>à</i>	<i>da</i>
	<i>he</i>	<i>il</i>	<i>er</i>



Characteristics of Current Corpora

Large:

- The Bank of English (Collins and University of Birmingham) from the 1990s contained more than 500 million words.
- The Colossal Clean Crawled Corpus (2020) contains 156 billion English tokens.

Available in many languages.

Easy to collect: the web is the largest corpus ever built and is accessible with a single mouse click.

Parallel corpora contain the same text in two or more languages (e.g., English/French Canadian Hansards, European Parliament proceedings in 23 languages).

Annotated with part-of-speech tags or manually parsed (treebanks):

- Characteristics/N of/PREP Current/ADJ Corpora/N
- (NP (NP Characteristics) (PP of (NP Current Corpora)))



Lexicography

Writing dictionaries.

Dictionaries for language learners should be based on real usage:

- *They're just trying to score **brownie points** with politicians*
- *The boss is pleased — that's another **brownie point***

Bank of English: *brownie point* (6 occurrences), *brownie points* (76 occurrences).

Extensive use of corpora to:

- Find **concordances** and cite real examples
- Extract **collocations** and describe frequent word pairs



Concordances

A word and its surrounding context:

Language	Concordances
English	s beginning of miracles did Je n they saw the miracles which n can do these miracles that t ain the second miracle that Je e they saw his miracles which
French	le premier des miracles que fi i dirent: Quel miracle nous mo om, voyant les miracles qu'il peut faire ces miracles que tu s ne voyez des miracles et des



Collocations

Word preferences: words that tend to occur together.

	English	French	German
You say	<i>Strong tea</i>	<i>Thé fort</i>	<i>Schmales Gesicht</i>
	<i>Powerful computer</i>	<i>Ordinateur puissant</i>	<i>Enge Kleidung</i>
You don't say	<i>Strong computer</i>	<i>Thé puissant</i>	<i>Schmale Kleidung</i>
	<i>Powerful tea</i>	<i>Ordinateur fort</i>	<i>Enges Gesicht</i>



Word Preferences

Strong <i>w</i>			Powerful <i>w</i>		
<i>strong w</i>	<i>powerful w</i>	<i>w</i>	<i>strong w</i>	<i>powerful w</i>	<i>w</i>
161	0	showing	1	32	than
175	2	support	1	32	figure
106	0	defense	3	31	minority
...					



Corpora as Knowledge Sources

Traditional use:

- Describe language usage more accurately

Machine learning:

- Learn statistical or machine-learning models for speech recognition, tagging, and parsing
- Evaluate tools such as part-of-speech taggers and parsers
- Derive models automatically from annotated or unannotated corpora

Applications:

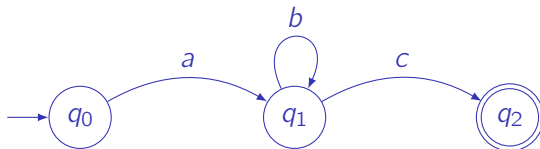
- Machine translation
- Information and knowledge extraction
- Question answering from textual sources
- Dialogue systems



Finite-State Automata

A flexible tool for searching and processing text.

A finite-state automaton (FSA) accepts and generates strings such as *ac*, *abc*, *abbc*, *abbbbc*, *abbbbbbbbbbbbc*, etc.



Finite-State Automata

Mathematically defined by:

- Q : a finite set of states
- Σ : a finite set of symbols or characters (the input alphabet)
- q_0 : a start state
- F : a set of final states, $F \subseteq Q$
- δ : a transition function $Q \times \Sigma \rightarrow Q$, where $\delta(q, i)$ returns the state reached when the automaton is in state q and consumes the input symbol i



Regular Expressions

Regular expressions (regexes) are equivalent to finite-state automata and are generally easier to use.

❶ Constant regular expressions:

Pattern	String
<code>regular</code>	<i>A section on <u>regular</u> expressions</i>
<code>the</code>	<i>The book of <u>the</u> life</i>

❷ Metacharacters such as `*`. The automaton above corresponds to the regular expression `ab*c`.

```
$ grep 'ab*c' myFile1 myFile2
```

Although `grep` was the first widely used regex tool, most programming languages now follow the Perl syntax.



Main Regex Operations

The two main regular-expression operations are matching and substitution.

They are often abbreviated using the Perl-style notation:

- The construct `m/regex/` denotes a match operation with the regular expression `regex`.
- The construct `s/regex/replacement/` performs a substitution: it matches the first occurrence of `regex` and replaces it with the `replacement` string.
- Modifiers can be added. For example, the `g` modifier (global) replaces all occurrences: `s/regex/replacement/g`.



regex101.com

regex101.com: a useful website for experimenting with and testing regular expressions.

The screenshot displays the regex101.com website interface. The top navigation bar includes the site logo, a search icon, and a settings icon. The main content area is divided into several sections:

- REGULAR EXPRESSION:** The input field contains the pattern `/ac?e/g`.
- TEST STRING:** The input field contains the text "The aerial acceleration alerted the ace pilot".
- EXPLANATION:** This section provides details about the matches:
 - `/ac?e/g`:
 - `a` matches the character `a` literally (case sensitive)
 - `c?` matches the character `c` literally (case sensitive)
 - Quantifier:** `*` Between zero and
- MATCH INFORMATION:** This section states: "No match groups were extracted. This means that your pattern matches but there were no (capturing groups) in it that matched anything in the subject string."
- QUICK REFERENCE:** This section includes a "FULL REFERENCE" link and a "MOST USED TOKENS" list:
 - most use...**: A single character of... [abc]
 - all tokens**: A character except... [^abc]
 - CATEGORIES**: A character in the class... [a-z]
- SUBSTITUTION:** This section is currently empty.

Metacharacters

Chars	Description	Examples
*	Matches any number of occurrences of the preceding character (zero or more)	ac*e matches ae, ace, acce, accce, etc., as in “The <u>a</u> erial <u>ac</u> celeration alerted the <u>ac</u> e pilot”
?	Matches at most one occurrence of the preceding character (zero or one)	ac?e matches ae and ace as in “The <u>a</u> erial acceleration alerted the <u>ac</u> e pilot”
+	Matches one or more occurrences of the preceding character	ac+e matches ace, acce, accce, etc., as in “The aerial <u>ac</u> celeration alerted the <u>ac</u> e pilot”



Metacharacters

Chars	Description	Examples
<code>{n}</code>	Matches exactly n occurrences of the preceding character	<code>ac{2}e</code> matches <code>acce</code> as in “The aerial <u>ac</u> celeration alerted the ace pilot”
<code>{n,}</code>	Matches n or more occurrences of the preceding character	<code>ac{2,}e</code> matches <code>acce</code> , <code>accce</code> , etc.
<code>{n,m}</code>	Matches between n and m occurrences of the preceding character	<code>ac{2,4}e</code> matches <code>acce</code> , <code>accce</code> , and <code>acccece</code>

Literal occurrences of metacharacters must be escaped with a backslash (`\`).



The Dot Metacharacter

The dot `.` is a metacharacter that matches any single character except a newline.

`a.e` matches the strings *ale* and *ace* in:

The aerial acceleration alerted the ace pilot

as well as *age*, *ape*, *are*, *ate*, *awe*, *axe*, or *aae*, *aAe*, *abe*, *aBe*, *a1e*, etc.

`.*` matches any string of characters up to a newline.



The Longest Match

The previous slide does not specify the matching strategy.

Consider the string *aabbcb* and the regular expression *a+b**.

By default the matching engine is greedy: it matches as early and as many characters as possible, resulting in *aabb*.

This can be problematic. Consider the regular expression *.** and the phrase:

They match as early and as many characters as they can.

A lazy strategy can be obtained with the **?* quantifier: *.*?*, which yields:

They match as early and as many characters as they can.



Character Classes

[...] matches any character contained in the list.

[^...] matches any character not contained in the list.

[abc] matches one occurrence of either a, b, or c.

[^abc] matches one occurrence of any character that is not a, b, or c.

[ABCDEFGHIJKLMNOPQRSTUVWXYZ] matches one uppercase unaccented letter.

[0123456789] matches one digit.

[0123456789]+\.[0123456789]+ matches decimal numbers.

[Cc]omputer [Ss]cience matches Computer Science, computer Science, Computer science, and computer science.



Predefined Character Classes

Expr.	Description	Equivalent	Example
<code>\d</code>	Any digit	<code>[0-9]</code>	<code>A\dC</code> matches <code>A0C</code> , <code>A1C</code> , <code>A2C</code> , <code>A3C</code> , etc.
<code>\D</code>	Any nondigit	<code>[^0-9]</code>	
<code>\w</code>	Any letter, digit, or underscore	<code>[a-zA-Z0-9_]</code>	<code>1\w2</code> matches <code>1a2</code> , <code>1A2</code> , <code>1b2</code> , <code>1B2</code> , etc.
<code>\W</code>	Any nonword character	<code>[^\w]</code>	
<code>\s</code>	Any whitespace character	<code>[\t\n\r\f]</code>	
<code>\S</code>	Any non-whitespace character	<code>[^\s]</code>	



Nonprintable Symbols or Positions

Char.	Description	Example
<code>^</code>	Matches the start of a line	<code>^ab*c</code> matches <code>ac</code> , <code>abc</code> , <code>abbc</code> , etc. at the beginning of a line
<code>\$</code>	Matches the end of a line	<code>ab?c\$</code> matches <code>ac</code> and <code>abc</code> at the end of a line
<code>\b</code>	Matches word boundaries	<code>\babc</code> matches <code>abcd</code> but not <code>dabc</code> <code>bcd\b</code> matches <code>abcd</code> but not <code>abcde</code>
<code>\n</code>	Matches a newline	<code>a\nb</code> matches <code>a</code> followed by a newline followed by <code>b</code>
<code>\t</code>	Matches a tab character	

```
egrep '^[aeiou]*$' myFile
```



Union and Boolean Operators

Union is denoted by $|$: $a|b$ matches either a or b .

The expression $a|bc$ matches the strings a and bc , while $(a|b)c$ matches ac and bc .

Order of precedence:

- 1 Closure and other repetition operators (highest)
- 2 Concatenation, line and word boundaries
- 3 Union (lowest)

abc^* denotes the set $\{ab, abc, abcc, abccc, \dots\}$.

$(abc)^*$ denotes $\{abc, abcabc, abcabcabc, \dots\}$.



Python

Match: m/regex/

```
import regex as re
line = 'The aerial acceleration alerted the ace pilot'
match = re.search('ab*c', line)
match # <regex.Match object; span=(11, 13), match='ac'>
match.group() # ac
```

The `re.search()` function stops at the first match.



Python

Use `findall()` or `finditer()` to obtain all matches.

Match: `m/regex/g`

```
match_list = re.findall('ab*c', line) # ['ac', 'ac']
```

Match: `m/regex/g`

```
match_iter = re.finditer('ab*c', line)
list(match_iter)
# [<regex.Match object; span=(11, 13), match='ac'>,
#  <regex.Match object; span=(36, 38), match='ac'>]
```



Match Modifiers

Flags that modify the matching behavior.

These flags are equivalent to Perl's `m/regex/modifiers`.

- Case-insensitive: `i`. The instruction `m/regex/i`. In Python this corresponds to the flag `re.I`.
- Multiline: `m` (`re.M` in Python). `m/regex/m`.
- Single-line (dot matches newline): `s` (`re.S` in Python). With the `/s` modifier, a dot in `m/regex/s` matches any character including newlines.

In Python, flags are specified as a sequence separated by vertical bars (`|`).



Python

Match: m/regex/modifiers

```
text = sys.stdin.read()
match = re.search('^ab*c', text, re.I | re.M) # m/^ab*c/im
if match:
    print('-> ' + match.group())
```



Python

Substitute: s/regex/replacement/g

```
for line in sys.stdin:
    if re.search('ab+c', line):
        print("Old: " + line, end='')
        # Replaces all occurrences
        line = re.sub('ab+c', 'ABC', line) # s/ab+c/ABC/g
        print("New: " + line, end='')
```

Substitute: s/regex/replacement/

To replace only the first occurrence:

```
# Replaces the first occurrence
line = re.sub('ab+c', 'ABC', line, 1) # s/ab+c/ABC/
```

Backreferences

Parts of a matched pattern can be stored using backreferences. These parts are enclosed in parentheses in the regular expression, for example:

`^(.)(b+)c+`

- The first part, `(.)`, is stored in variable `\1`
- The second part, `(b+)`, is stored in `\2`, and so on.

Backreferences

```
line = 'abbbcdeeeef'
match = re.search('^(.)(b+)c+', line)
match.group(1) # 'a'
match.group(2) # 'bbb'
```



Python

Backreferences

The pattern `m/(.)\1\1/` matches sequences of three identical characters:

```
line = 'abbbcdeeef'  
match = re.search(r'(\1)\1\1', line)  
match.group(1) # 'b'
```

A raw string (prefix `r`) must be used; otherwise `\1` would be interpreted as an octal number.

Substitutions

```
s/(.)\1\1/***/g  
  
re.sub(r'(\1)\1\1', '***', 'abbbcdeeef') # 'a***cd***f'
```

Python

Multiple backreferences

Python can create as many capture groups as needed: \1, \2, \3, etc. Outside the regular expression these groups are accessed via `group(<digit>)`: `match_object.group(1)`, `match_object.group(2)`, `match_object.group(3)`, etc.

Multiple backreferences

```
m/\$ *([0-9]+)\.?([0-9]*)/
```

```
price = "We'll buy it for $72.40"
```

```
match = re.search(r'\$ *([0-9]+)\.?([0-9]*)', price)
```

```
match.group() # '$72.40'    The entire match
```

```
match.group(1) # '72'       The first group
```

```
match.group(2) # '40'       The second group
```

Python

Substitutions

```
s/\$ *([0-9]+)\.?([0-9]*)/\1 dollars and \2 cents/g

price = "We'll buy it for $72.40"
re.sub(r'\$ *([0-9]+)\.?([0-9]*)',
      r'\1 dollars and \2 cents', price)
# We'll buy it for 72 dollars and 40 cents
```



Python

Match objects

- `match_object.group()` or `match_object.group(0)` returns the entire match.
- `match_object.group(n)` returns the n -th parenthesized subgroup.

In addition, `match_object.groups()` returns a tuple containing all groups, and `match_object.string` contains the original input string.

```
price = "We'll buy it for $72.40"  
match = re.search(r'\$ *([0-9]+)\.?([0-9]*)', price)  
match.string    # We'll buy it for $72.40  
match.groups() # ('72', '40')
```



Python

Match objects

The start and end indices of matched substrings can be obtained with:

```
match_object.start([group])  
match_object.end([group])
```

```
line = """Tell me, O muse, of that ingenious hero  
    who travelled far and wide after he had sacked  
    the famous town of Troy."""
```

```
match = re.search(',.*', line, re.S)  
line[0:match.start()]           # 'Tell me'  
line[match.start():match.end()] # ', O muse,'  
line[match.end():]              # 'of that ingenious hero  
                                # who travelled far and wide a  
                                # the famous town of Troy.'
```

A Regex to Find Concordances

To print concordances we need a regular expression that matches the pattern together with a left and right context.

For example, *Nils Holgersson* with a context of 15 characters:

```
.{0,15}Nils Holgersson.{0,15}
```

Ideally the pattern and the context width are passed as parameters:

```
pattern = 'Nils Holgersson'  
width = 15  
'.{0,width}pattern.{0,width}'
```



format()

`str.format()` provides variable substitution:

```
begin = 'my'
'{} string {}'.format(begin, 'is empty')
# 'my string is empty'
```

`format()` offers many options, including argument reordering via indices:

```
begin = 'my'
'{1} string {0}'.format('is empty', begin)
# 'my string is empty'
```

If the format string itself contains braces, they must be escaped by doubling them: `{{` for a literal `{` and `}}` for `}`.

```
('.{0},{width}}{pattern}.{0},{width}}}'
    .format(pattern=pattern, width=width))
```



Concordances in Python

```
[file_name, pattern, width] = sys.argv[1:]
try:
    text = open(file_name).read()
except:
    print('Could not open file', file_name)
    exit(0)

# Spaces match tabs and newlines
pattern = re.sub(' ', r'\s+', pattern)
# Replace newlines with spaces in the text
text = re.sub(r'\s+', ' ', text)
concordance = ('({0,{width}}){pattern}.({0,{width}})'.
               .format(pattern=pattern, width=width))
for match in re.finditer(concordance, text):
    print(match.group(1))
```



Approximate String Matching

A set of edit operations that transform a source string into a target string: copy, substitution, insertion, deletion, and transposition (reversal). Edit examples for *acress* from Kernighan et al. (1990):

Type	Correction	Source	Target	Position	Operation
acress	actress	—	t	2	Insertion
acress	cress	a	—	0	Deletion
acress	caress	ac	ca	0	Transposition
acress	access	r	c	2	Substitution
acress	across	e	o	3	Substitution
acress	acres	s	—	4	Deletion
acress	acres	s	—	5	Deletion



Building a Spell Checker

Spell checkers use a dictionary together with a set of transformations to suggest corrections for misspelled words.

Dictionaries are typically collected from well-written texts (novels, newspapers, etc.).

- Given a word that does not appear in the dictionary, the spell checker generates all single-edit transformations of that word.
- If only one edit operation is allowed on a source string of length n , and the alphabet consists of 26 unaccented letters:
 - deletion produces n new strings;
 - insertion produces $(n + 1) \times 26$ strings;
 - substitution produces $n \times 25$ strings;
 - transposition produces $n - 1$ new strings.
- The spell checker retains those transformations that exist in the dictionary and ranks them by frequency to suggest the most likely correction.



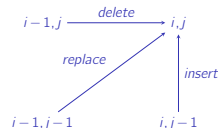
For an implementation see <http://norvig.com/spell-correct.html>

Building a Spell Checker

```
freq('acres') = 36  
freq('caress') = 3  
freq('cress') = false  
freq('actress') = 7  
freq('access') = 56  
freq('across') = 222
```



Distance between ab and cb



Edit distances measure the similarity between strings.

Consider the alignment

a	b	Source
c	b	Destination

b	2		
c	1		
Start	0	1	2
	Start	a	b



Minimum Edit Distance

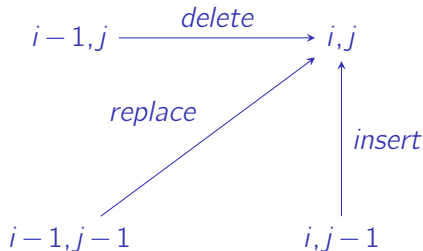
The minimum edit distance is computed using a matrix whose entry at position (i,j) is defined by the recurrence:

$$\text{edit_distance}(i,j) = \min \left\{ \begin{array}{l} \text{edit_distance}(i-1,j) + \textit{del_cost} \\ \text{edit_distance}(i-1,j-1) + \textit{subst_cost} \\ \text{edit_distance}(i,j-1) + \textit{ins_cost} \end{array} \right\}$$

with base cases $\text{edit_distance}(i,0) = i$ and $\text{edit_distance}(0,j) = j$.



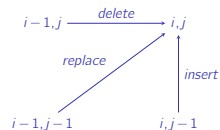
Edit Operations



Usually $del_cost = ins_cost = 1$,
 $subst_cost = 2$ if $source(i) \neq target(j)$,
 $subst_cost = 0$ if $source(i) = target(j)$.



Distance between ab and cb



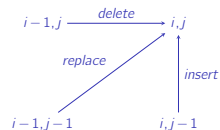
Alignment:

a	b	Source
c	b	Destination

b	2		
c	1	2	
Start	0	1	2
	Start	a	b



Distance between ab and cb



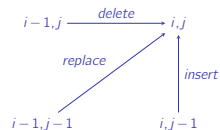
Alignment:

a	b	Source
c	b	Destination

b	2	3	
c	1	2	3
Start	0	1	2
	Start	a	b



Distance between ab and cb



Alignment:

a	b	Source
c	b	Destination

b	2	3	2
c	1	2	3
Start	0	1	2
Start	a	b	



Distance between *language* and *lineage*

e	7								
g	6								
a	5								
e	4								
n	3								
i	2								
l	1								
Start	0	1	2	3	4	5	6	7	8
	Start	l	a	n	g	u	a	g	e



Distance between *language* and *lineage*

e	7	6	5						
g	6	5	4						
a	5	4	3						
e	4	3	4						
n	3	2	3						
i	2	1	2	3	4	5	6	7	8
l	1	0	1	2	3	4	5	6	7
Start	0	1	2	3	4	5	6	7	8
	Start	l	a	n	g	u	a	g	e



Distance between *language* and *lineage*

e	7	6	5	6	5	6	7	6	5
g	6	5	4	5	4	5	6	5	6
a	5	4	3	4	5	6	5	6	7
e	4	3	4	3	4	5	6	7	6
n	3	2	3	2	3	4	5	6	7
i	2	1	2	3	4	5	6	7	8
l	1	0	1	2	3	4	5	6	7
Start	0	1	2	3	4	5	6	7	8
	Start	l	a	n	g	u	a	g	e



Python Code

```
[source, target] = sys.argv[1:]  
length_s = len(source) + 1  
length_t = len(target) + 1  
# Initialize first row and column  
table = [None] * length_s  
for i in range(length_s):  
    table[i] = [None] * length_t  
    table[i][0] = i  
for j in range(length_t):  
    table[0][j] = j
```



Python Code

```
# Fill the table. Row and column indices start at 1
for i in range(1, length_s):
    for j in range(1, length_t):
        # Copy or substitution?
        cost = 0 if source[i-1] == target[j-1] else 2
        # Compute the minimum
        minimum = table[i-1][j-1] + cost
        if minimum > table[i][j-1] + 1:
            minimum = table[i][j-1] + 1
        if minimum > table[i-1][j] + 1:
            minimum = table[i-1][j] + 1
        table[i][j] = minimum
print('Minimum distance:', table[length_s-1][length_t-1])
```



Distance between *language* and *lineage*

	First alignment	Third alignment
Without epsilon symbols	l a n g u a g e	l a n g u a g e
	/ / /	/ / /
	l i n e a g e	l i n e a g e
With epsilon symbols	l a n g u a g e	l a n g u ε a g e
	l i n e ε a g e	l i n ε ε e a g e

